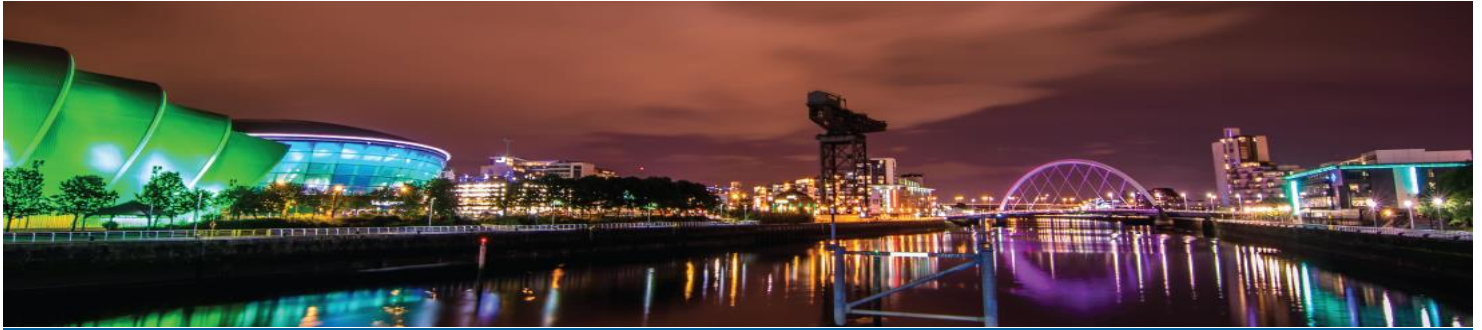




CALL FOR PAPERS: Optical Networks & Systems Symposium

Symposium Co-Chairs

- Chedlia Ben Naila, Nagoya University, Japan, bennaila@ieee.org
- Iman Tavakkolnia, University of Cambridge, United Kingdom, it360@cam.ac.uk

SCOPE AND MOTIVATION

Communication networks are rapidly evolving to support next-generation services that will significantly impact our daily lives. These services include autonomous driving, holographic video, immersive and extended reality technologies, smart environments, metaverse applications, and massive-scale connectivity for billions of sensors, drones, robots, smart objects, wearables, and other intelligent devices. The delivery of these services will introduce unprecedented challenges for network operators, who must ensure their infrastructure can address stringent demands related to bandwidth, latency, energy efficiency, ubiquitous coverage, availability, and resilience. Moreover, future networks must be adaptive and intelligent, with the capability to meet the requirements for highly diverse services. For instance, the network must be able to provide stable connectivity to a few users in a remote rural area and simultaneously support tens of thousands of users densely packed into a stadium in an urban setting. Moreover, **energy efficiency** remains a central challenge and motivator in the design and operation of future optical networks. **Reducing power consumption without compromising performance is a key objective**, driving innovation in low-power transceivers, energy-aware network protocols, and intelligent dynamic energy management strategies.

In this context, **optical networks and systems** will play a foundational role in realizing these demands. They offer a sustainable approach from both **cost and energy efficiency standpoints** and must integrate seamlessly with both radio and non-terrestrial networks. Optical systems are expected to support all segments of the communication infrastructure, from the core network to metro networks, fixed and **wireless optical access**, and even intra-data centre networks. **Wired and wireless optical communication technologies are crucial for enabling ultra-high-capacity, low-latency connectivity across diverse deployment scenarios, ensuring optimised end-to-end performance for future applications.** As such, they form the backbone of a flexible, scalable, and high-performance network infrastructure for 6G and beyond.

Numerous technological breakthroughs in the field of optical communication are currently under investigation, and several have already demonstrated maturity. Nevertheless, many open challenges remain. In the realm of optical fibre systems, experiments with hollow-core fibres have shown the potential to match or even exceed the performance over conventional fibres, while offering reduced attenuation and propagation delays. However, robust physical layer models and viable long-distance deployment strategies are still under active research. Meanwhile, the use of extended wavelength bands—such as the C+L+S bands has proven effective in increasing overall network throughput. This advancement, however, requires the development of specialized optical components such as broadband Ytterbium, Praseodymium, or Erbium-doped fibre amplifiers (Y/P/EDFAs), tuneable transceivers, and wideband optical filters to operate efficiently across extended spectral bands, which are not fully supported by conventional equipment.

In parallel, spatial division multiplexing (SDM) techniques, such as multi-core and multi-fibre architectures, are being developed to further boost the capacity of optical fibres. These approaches also require accurate modelling at the physical layer to support effective multilayer network optimization. Quality of Transmission (QoT) estimation models will be critical in solving classical and extended networking challenges, such as Routing and Spectrum Assignment (RSA).

On another front, the convergence of optical networking with emerging data-centric technologies is opening new dimensions. The adoption of data analytics, artificial **intelligence (AI)**, **machine learning (ML)**, and **large language models (LLMs)**, alongside next-generation networking paradigms like Software-Defined Networking (SDN), Open Radio Access Networks (ORAN), Network Function Virtualization (NFV), cloud and edge computing, in-network computing, telemetry, and **digital twin technologies**, is expected to usher in a new era of autonomous optical network. This environment will foster the realization of “Zero-X” networks, where operations are zero-touch, zero-wait, and zero-trouble, enabling seamless orchestration, control, and management.

Quantum optical networks are another critical area of ongoing development. With the promise of ultra-secure communications enabled by quantum key distribution (QKD), entanglement-based protocols, and the integration of quantum memory and repeater technologies, quantum-enhanced optical networks will form the bedrock for future quantum-secure communication infrastructures, offering unprecedented security guarantees and capabilities.

Architecturally, the integration between terrestrial optical networks and non-terrestrial infrastructures, such as Low-Earth-Orbit (LEO) satellite mega-constellations, is poised to redefine global Internet access. The combined strengths of these two network types, high-throughput, low-latency terrestrial fibre and the expansive reach of spaceborne networks, could enable ubiquitous and resilient global connectivity. Essential to this integration will be robust ground station connectivity, optical inter-satellite links, and unified management frameworks.

The development of free-space optical (FSO) systems, hybrid radio-FSO architectures, and **optical wireless communication (OWC) technologies is gaining significant momentum**, supported by emerging standards aimed at enabling their seamless **integration into future network infrastructures**. These technologies offer indoor multiuser mobile access connectivity, such as Light Fidelity (LiFi). Several OWC technologies, such as visible light communication (VLC), infrared communication, and ultraviolet (UV) communications expand the potential use cases across challenging environments, including dense urban landscapes, inter-satellite links unmanned aerial vehicle (UAV) networks, and underwater communication scenarios where traditional RF-based solutions are limited or ineffective. By **combining radio and optical wireless technologies, hybrid systems** can provide enhanced resilience, improved spectral efficiency, and increased deployment flexibility. **In addition to communication, optical wireless technologies are expanding their role into sensing, localization, and energy delivery. Optical wireless sensing and positioning techniques offer precise environmental awareness, which are critical for applications in future networks, smart buildings, autonomous navigation, and industrial automation.** Furthermore, **optical wireless power transfer is emerging as a promising solution to enable efficient energy delivery to distributed units.**

Finally, optical and electro-optical systems will not only enhance traditional communication capacities to meet the demands of future networks but will also serve as the foundation for novel architectures and technologies. Advanced optical interconnects are expected to reshape transport and access networks, particularly in support of future antenna systems. Furthermore, the development and deployment of photonic integrated circuits (PICs) promise significant gains in terms of reducing size, weight, manufacturing cost, and power consumption, while increasing reliability. These innovations will support a wide range of applications, from photonic neural networks to lab-on-chip biosensing technologies, paving the way for a truly intelligent, sustainable, and interconnected future.

The Optical Networks and Systems Symposium aims to bring together researchers, practitioners, and technologists to exchange ideas and stimulate further research and developments in this exciting era for the optical network infrastructure in the future hyperconnected world.

Submitted papers should be between **4 and 6 pages in length**. Paper acceptance decisions will be based on novelty of the work, quality of results and clarity of the presentation.

CALL FOR PAPERS: Optical Networks & Systems Symposium

TOPICS OF INTEREST

The Optical Networks and Systems Symposium intends to showcase the latest developments in all research areas related to optical networks and systems. The Symposium cordially invites original contributions in, but not limited to, the following topical areas and others not explicitly listed but that are closely related:

- Artificial intelligence and machine learning for optical systems and networks
- Coding, modulation, and signal processing for optical systems
- Cross-layer design of optical networks
- Data analytics for optical networks
- Digital twin in optical networks and streaming telemetry
- Energy-efficiency of optical networks
- Experimental data-driven optical networking
- Free-space optical (FSO) communications
- High quality service provisioning for multimedia applications
- Hybrid fibre-coax (HFC) networks
- Integrated optical communication and sensing
- Inter-satellite optical communication links
- Multiple access techniques for mobile optical wireless systems
- Optical access systems & networks in support of cost-effective edge computing deployment
- Optical and wireless network convergence and mobile x-haul
- Optical camera communications (OCC)
- Optical interconnects for datacentres & high-performance computing
- Optical network architectures, design, and performance evaluation
- Optical network control and management
- Optical network for inter- and intra-datacentre connectivity
- Optical network security
- Optical network survivability and availability
- Optical network testbeds and experiments
- Optical wireless communication (OWC) and LiFi
- Optical wireless positioning systems
- Optical wireless vehicular networks
- Photonic integrated circuits for optical communication networks
- Photonic neural networks
- Quantum communication systems and networking
- Radio-over-fibre and radio over FSO systems
- Satellite optical communication systems & networking (Satellite-satellite, Satellite-Ground)
- Standardization in optical networks including optical wireless
- Software-defined optical networks including programmability, automation, and disaggregation
- Systems & networks for open and disaggregated optical transport
- Underwater optical communications
- Ultraviolet communications

BIOGRAPHIES OF CO-CHAIRS

Chedlia Ben Naila received the Ph.D. degree from Waseda University, Tokyo, Japan, in 2012. She is currently an Assistant Professor at Nagoya University, Nagoya, Japan. Previously, she was a Junior Lecturer at Iset'Com, Tunisia (2005–2008), and as an Adjunct Lecturer at Waseda University, Japan (2014–2019). She is a Senior member of IEEE and member of OPTICA (formerly OSA). She was the recipient of the 2011 Waseda University Ono Azusa Memorial Academic Award medal, the Best Paper Award of IEEE Journal (2012), the 26th Telecom System Technology Award from the Japanese Telecommunication Advancement Foundation (TAF) of Japan (2011), and the IEEE ITU-T Kaleidoscope Best Paper award (2011). Her current research interests include optical wireless communications (OWC), free-space optics (FSO) systems, optical intelligent reflecting surfaces (IRS), and underwater communications. She is also an organizer of several workshops for IEEE VTC, ISCC, etc.

Iman Tavakkolnia received the Ph.D. degree from The University of Edinburgh, in 2018. He is currently an Assistant Professor with the Electrical Engineering Division, an academic member of the LiFi Research and Development Centre, and a Fellow of Selwyn College, University of Cambridge. His research interests include developing a fundamental understanding of the energy efficiency of current and future telecommunication systems and lies on the frontier of communication theory, advanced materials, signal processing, and optical communications. He is a Co-Investigator on two EPSRC Future Communication Hubs in the UK (TITAN and HASC) as well as project REASON under the Future Open Networks Research Challenge grant funded by the UK's Department of Science, Innovation and Technology. He is an editor of IEEE Transactions on Communications, associate editor of IEEE Communication Letters, and the organiser of several workshops for IEEE WCNC, INFOCOM, and GLOBECOM.

HOW TO SUBMIT A PAPER

All papers for technical symposia should be submitted via EDAS. Full instructions on how to submit papers and important deadlines are posted at <https://icc2026.ieee-icc.org/>

The authors of selected papers from this symposium will be invited to submit an extended version of their work for fast-track review and possible publication in the IEEE Open Journal of the Communications Society.

IMPORTANT DATES

Paper Submission: 29 September 2025

Acceptance Notification: 12 January 2026

Camera-Ready : 13 February 2026

<http://icc2026.ieee-icc.org>